Assignment 2 - Math

TW Lab 2025

Write LaTeX code to output the following mathematical expressions using appropriate mathematical environments and references.

1 Basics

This famous Einstein's mass-energy equation, $E = m.c^2$ should be rendered inline.

However the following equation of motion should be rendered in a block:

$$v = u + a.t$$

2 Fractions & Roots

$$x_i = \frac{\sqrt{a^2 + b_i^2}}{c_i}$$

3 Symbols & Math-fonts

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}, \quad \forall x \in \mathbb{R}\ \exists \epsilon > 0 \Rightarrow |x| < \epsilon$$

$$F(\theta) = \sum_{k=0}^{n} \binom{n}{k} \theta^k (1-\theta)^{n-k}$$

4 Calculus

$$\frac{du}{dx} + \frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}$$

$$\int_0^1 (3x^2 + 2x + 1)\, dx$$

5 Multi-line equations

5.1 Centering & alignment

$$x = 5 \tag{1}$$
$$x + y = 3 \tag{2}$$

$$x = 5 \tag{3}$$
$$x + y = 3 \tag{4}$$

5.2 Multiline equation

$$\begin{aligned}(x+y+z)^3 &= (x+y+z).(x+y+z).(x+y+z) \\ &= x^3+y^3+z^3+3x^2y+3xy^2 \\ &\quad +3y^2z+3yz^2+3x^2z+3xz^2+6xyz\end{aligned}$$

6 Matrices

$$\begin{matrix}1 & 2\\ 3 & 4\end{matrix} \qquad \begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix} \qquad \begin{vmatrix}1 & 2\\ 3 & 4\end{vmatrix} \qquad \begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}$$